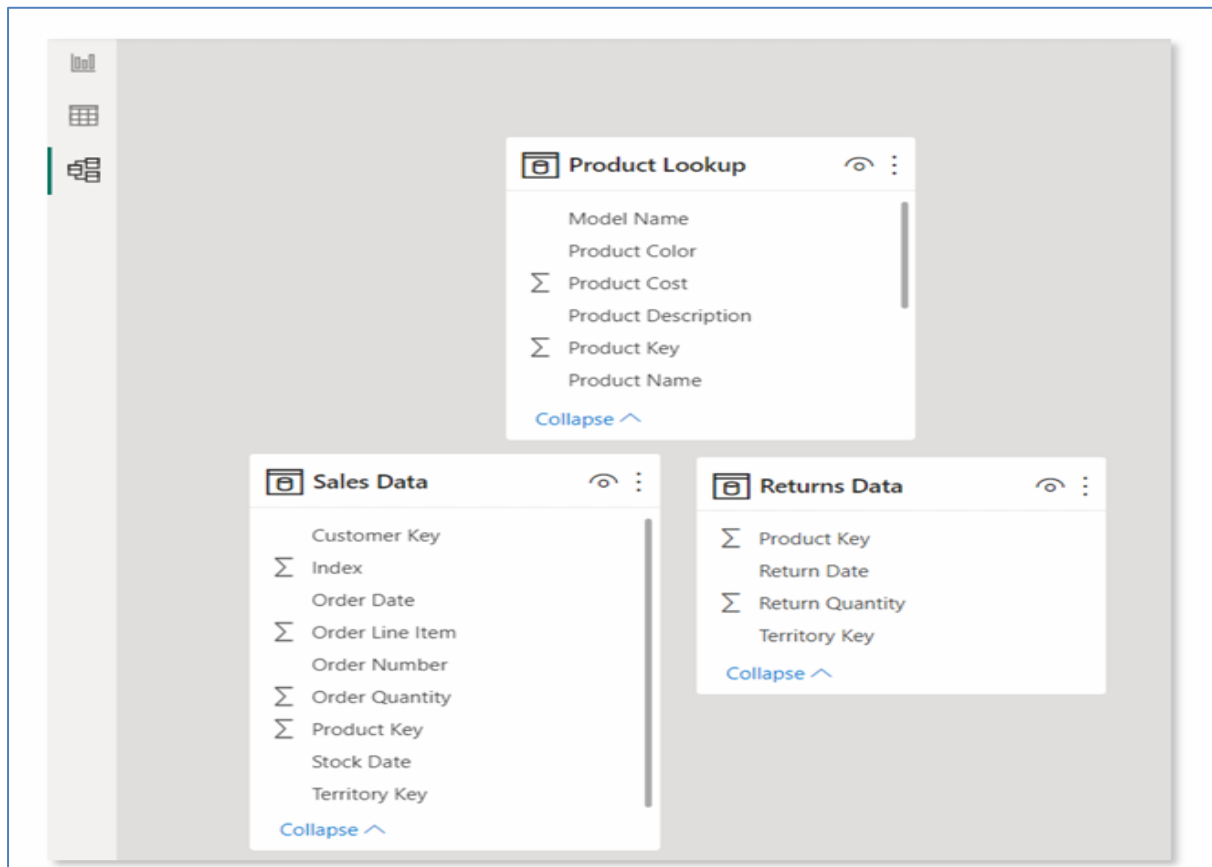


CREATING A DATA MODEL

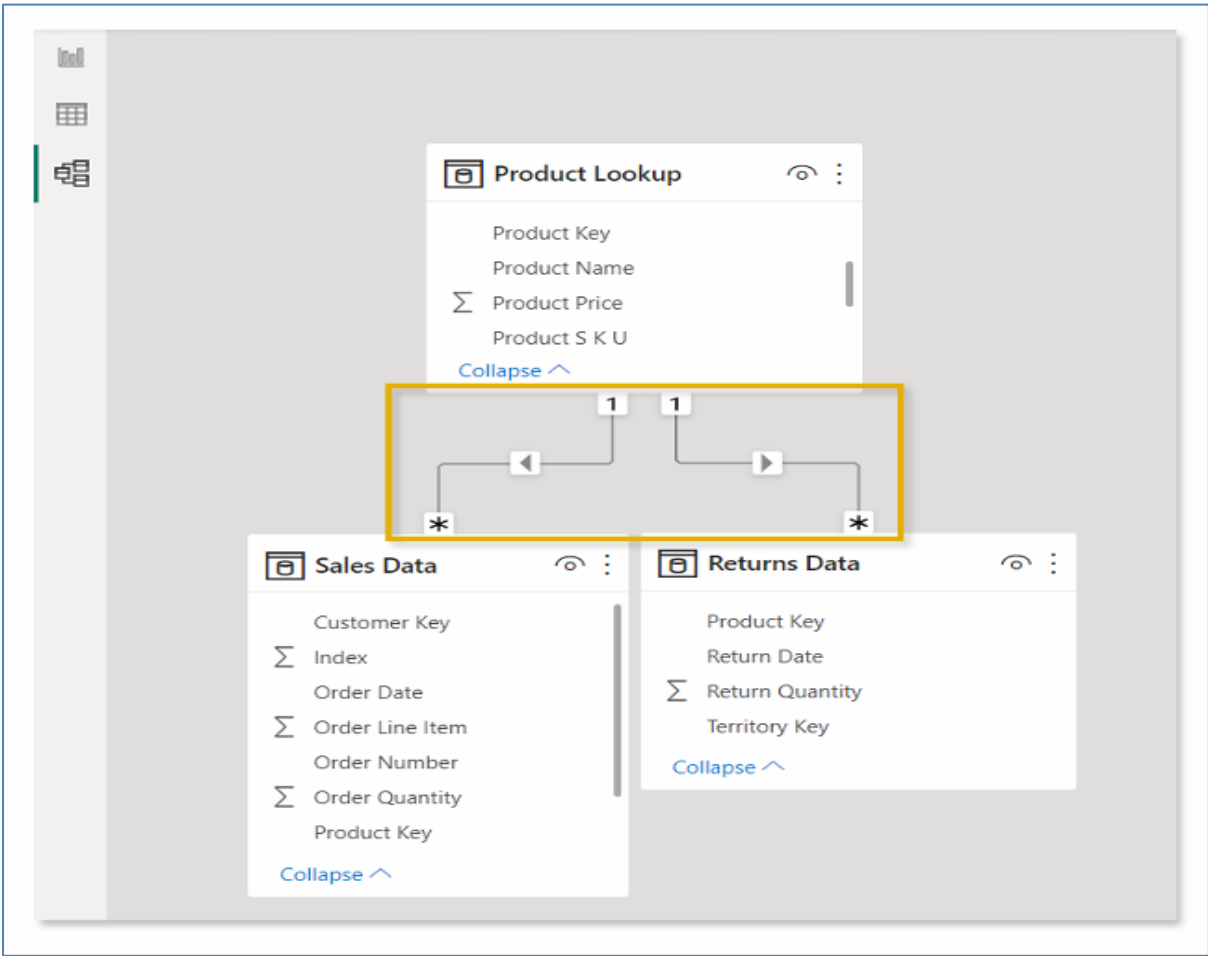
WHAT IS A DATA MODEL?

This is a collection of independent tables, which share no connections or relationships.



ProductName	OrderQuantity	ReturnQuantity
All-Purpose Bike Stand	84,174	1,828
AWC Logo Cap	84,174	1,828
Bike Wash - Dissolver	84,174	1,828
Cable Lock	84,174	1,828
Chain	84,174	1,828
Classic Vest, L	84,174	1,828
Classic Vest, M	84,174	1,828
Classic Vest, S	84,174	1,828
Fender Set - Mountain	84,174	1,828
Total	84,174	1,828

The tables are connected via relationships, based on a common field (Product Key)



ProductName	OrderQuantity	ReturnQuantity
All-Purpose Bike Stand	234	8
AWC Logo Cap	4,151	46
Bike Wash - Dissolver	1,706	25
Classic Vest, L	182	4
Classic Vest, M	182	7
Classic Vest, S	157	8
Fender Set - Mountain	3,960	54
Half-Finger Gloves, L	840	18
Half-Finger Gloves, M	918	16
Total	84,174	1,828

DATABASE NORMALIZATION

Eliminate redundant data to decrease table sizes and improve processing speed & efficiency

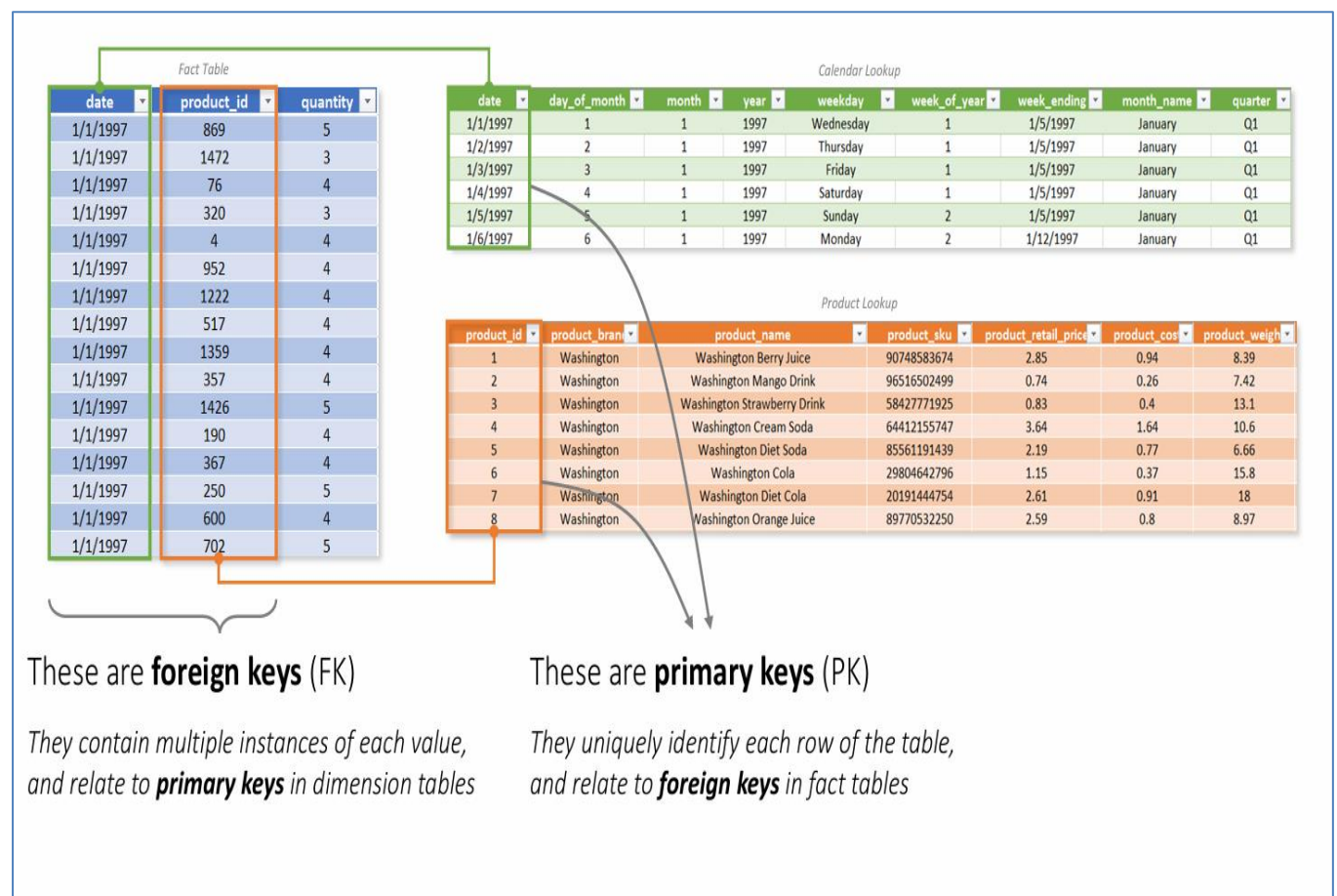
- Minimize errors and anomalies from data modifications (inserting, updating or deleting records)
- Simplify queries and structure the database for meaningful analysis

FACT & DIMENSION TABLES

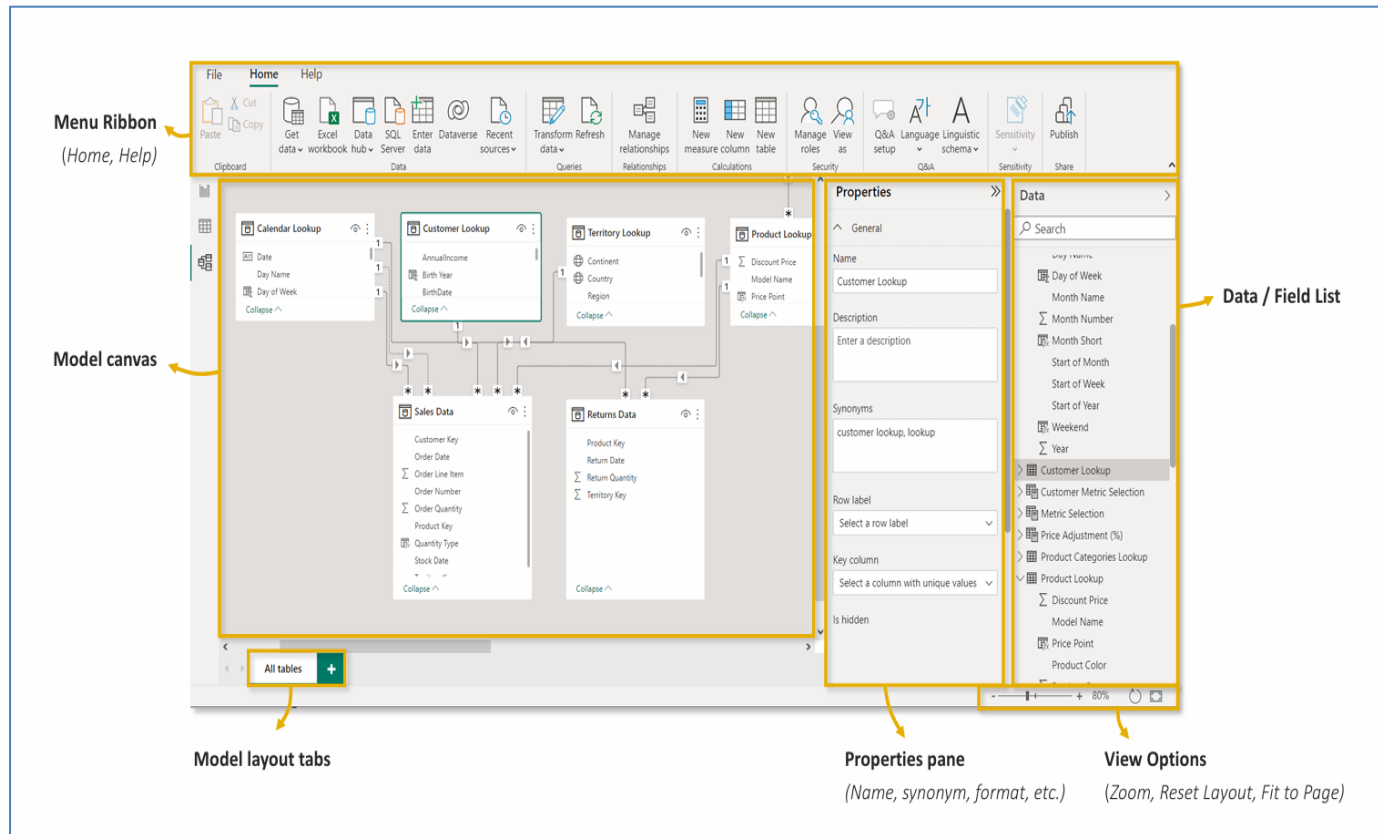
Data models generally contain two types of tables: fact (“**data**”) tables, and dimension (“**lookup**”) tables:

- Fact tables contain **numerical values** or metrics used for summarization (sales, orders, transactions, pageviews, etc.)
- Dimension tables contain **descriptive attributes** used for filtering or grouping (products, customers, dates, stores, etc.)

PRIMARY & FOREIGN KEYS

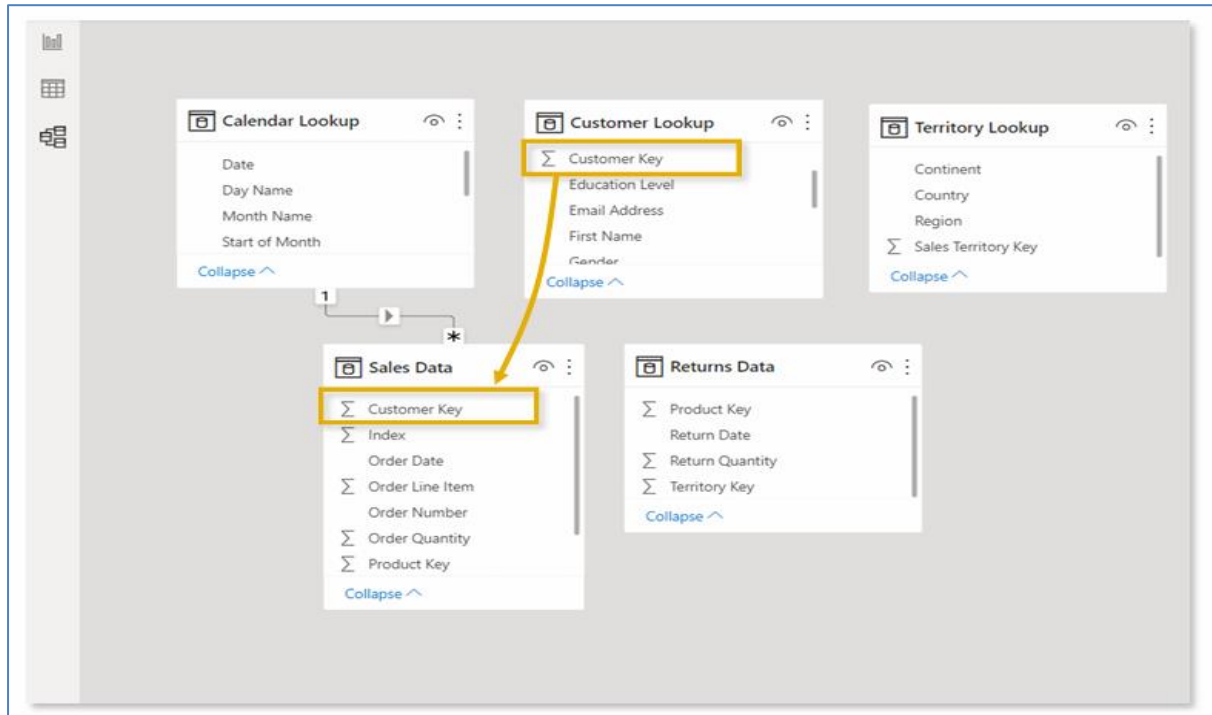


THE MODEL VIEW

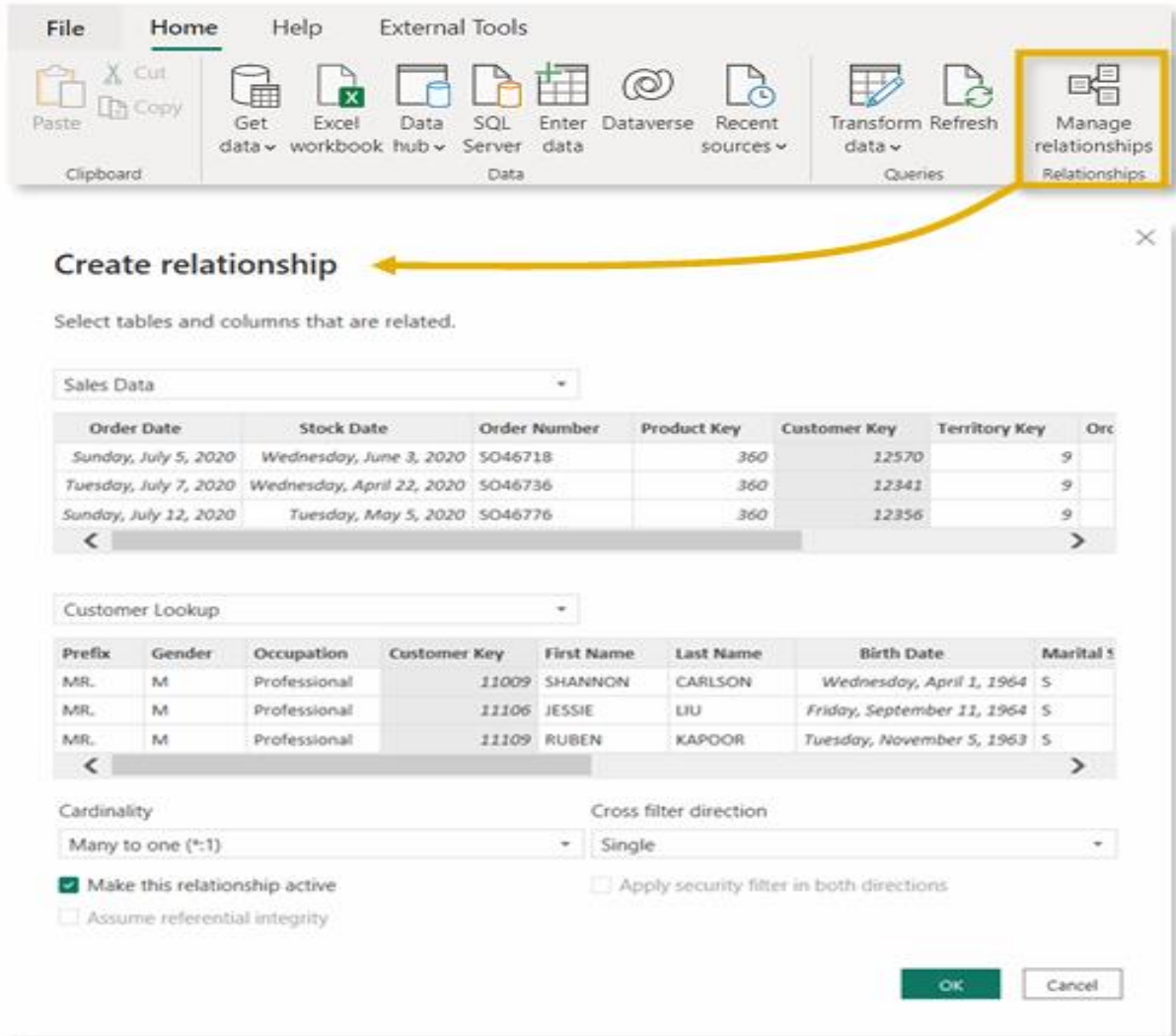


CREATING TABLE RELATIONSHIPS

OPTION 1: Click and drag to connect primary and foreign keys within the Model view



OPTION 2: Add or detect relationships using the Manage Relationships dialog box



The screenshot shows the Power BI Desktop ribbon with the 'Home' tab selected. The 'Manage relationships' button is highlighted in the 'Queries' group. A yellow arrow points from this button to the 'Create relationship' dialog box.

Create relationship

Select tables and columns that are related.

Sales Data

Order Date	Stock Date	Order Number	Product Key	Customer Key	Territory Key	Order
Sunday, July 5, 2020	Wednesday, June 3, 2020	SO46718	360	12570		9
Tuesday, July 7, 2020	Wednesday, April 22, 2020	SO46736	360	12341		9
Sunday, July 12, 2020	Tuesday, May 5, 2020	SO46776	360	12356		9

Customer Lookup

Prefix	Gender	Occupation	Customer Key	First Name	Last Name	Birth Date	Marital S
MR.	M	Professional	11009	SHANNON	CARLSON	Wednesday, April 1, 1964	S
MR.	M	Professional	11106	JESSIE	LIU	Friday, September 11, 1964	S
MR.	M	Professional	11109	RUBEN	KAPOOR	Tuesday, November 5, 1963	S

Cardinality: Many to one (*:1)

Cross filter direction: Single

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

OK Cancel

MANAGING & EDITING RELATIONSHIPS

Manage relationships

Active	From: Table (Column)	To: Table (Column)
☒	Product Lookup (Product Subcategory Key)	Product Subcategories Lookup (Product Subcategory Key)
☒	Product Subcategories Lookup (Product Category Key)	Product Categories Lookup (Product Category Key)
☒	Sales Data (Customer Key)	Customer Lookup (Customer Key)
☒	Sales Data (Order Date)	Calendar Lookup (Date)
☒	Sales Data (Product Key)	Product Lookup (Product Key)

New... Autodetect... **Edit...** Delete

Close

Edit relationship

Select tables and columns that are related.

Sales Data

Order Date	Stock Date	Order Number	Product Key	Customer Key	Territory Key	Order Date
Sunday, July 5, 2020	Wednesday, June 3, 2020	SO46718	360	12570	9	
Tuesday, July 7, 2020	Wednesday, April 22, 2020	SO46736	360	12341	9	
Sunday, July 12, 2020	Tuesday, May 5, 2020	SO46776	360	12356	9	

Customer Lookup

Prefix	Gender	Occupation	Customer Key	First Name	Last Name	Birth Date	Marital S
MR.	M	Professional	11009	SHANNON	CARLSON	Wednesday, April 1, 1964	S
MR.	M	Professional	11106	JESSIE	LIU	Friday, September 11, 1964	S
MR.	M	Professional	11109	RUBEN	KAPOOR	Tuesday, November 5, 1963	S

Cardinality: Many to one (*:1) Cross filter direction: Single

☒ Make this relationship active ☐ Apply security filter in both directions

☐ Assume referential integrity

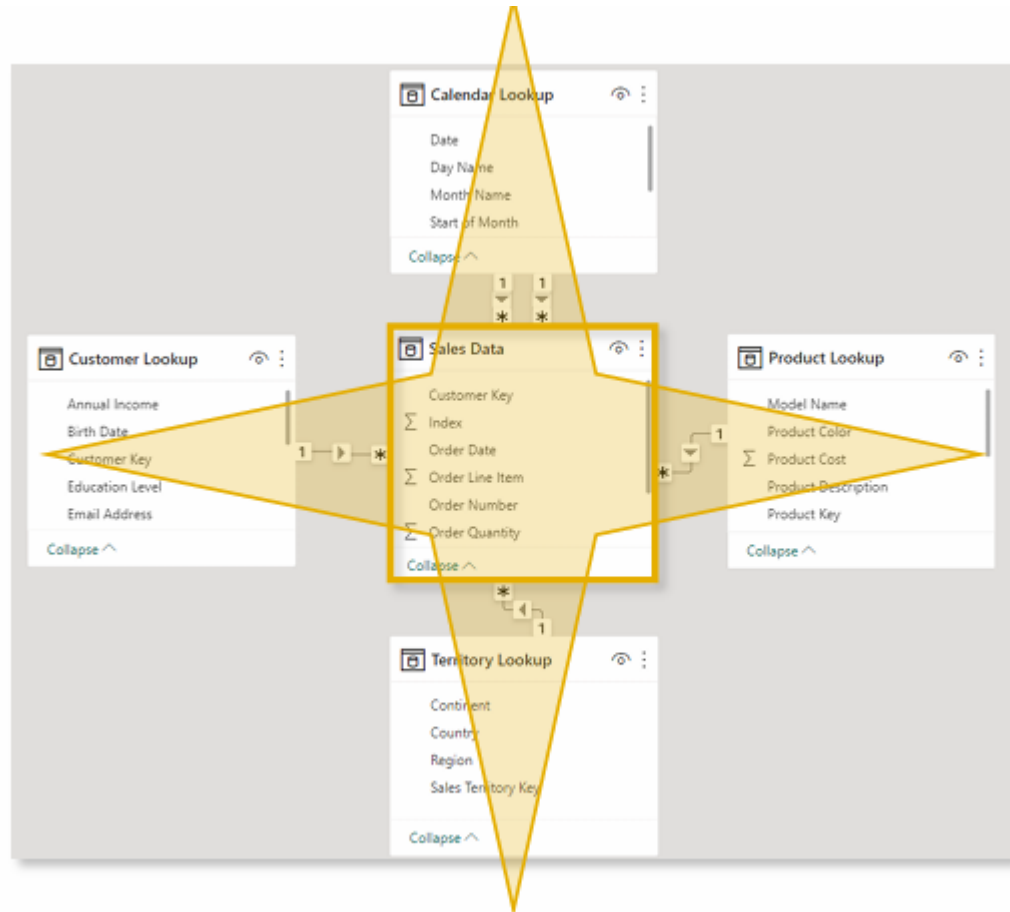
OK Cancel

Launch the **Manage Relationships** dialog box or double-click a relationship to modify it

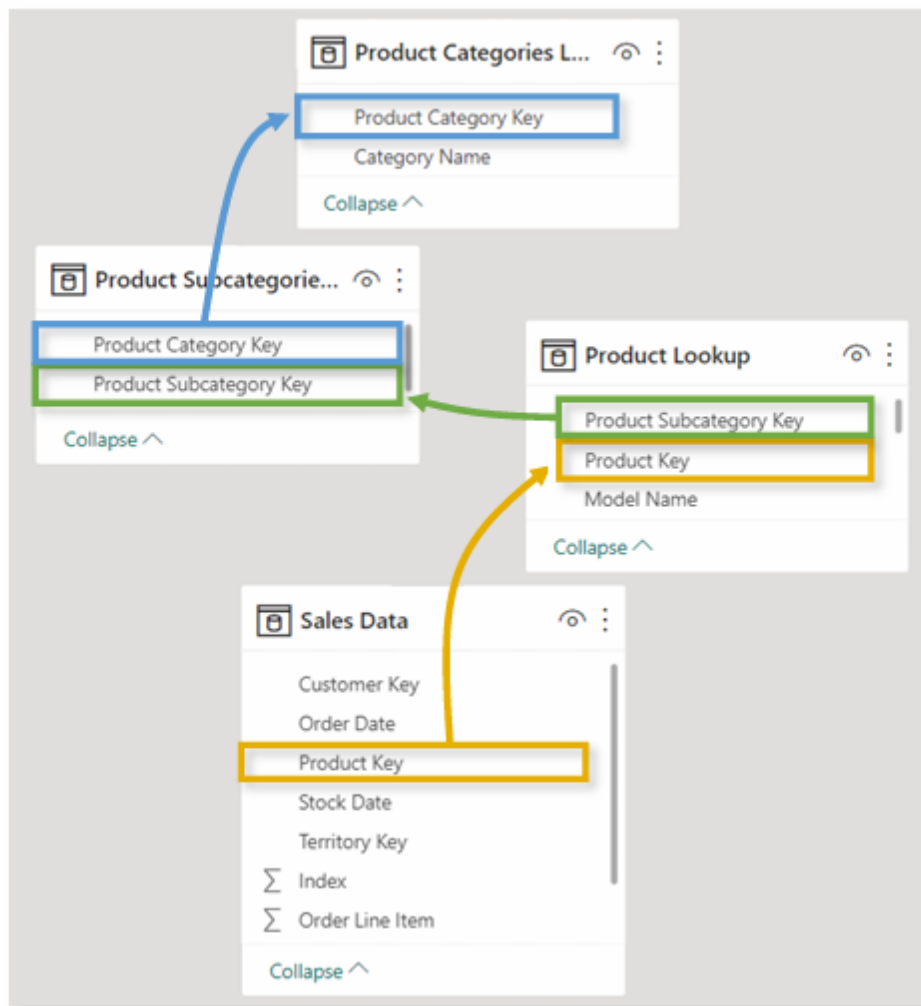
Editing tools allow you to **activate or deactivate** relationships and manage **cardinality** and **filter direction** – more on that soon!

STAR & SNOWFLAKE SCHEMAS

A **star** schema is the simplest and most common type of data model, characterized by a single fact table surrounded by related dimension tables



A **snowflake** schema is an extension of a star, and includes relationships between dimension tables and related sub-dimension tables



ACTIVE RELATIONSHIPS

The screenshot shows the Power BI interface with the 'Calendar ...' table selected. The 'Sales Data' table is also visible. A relationship is established between 'Order Date' in 'Sales Data' and 'Date' in 'Calendar ...'. The relationship is active, as indicated by the 'Make this relationship active' checkbox being checked in the 'Edit relationship' dialog.

Calendar ...

- Date
- Day Name
- Month Name
- Start of Month
- Start of Week
- Start of Year
- Σ Year

Sales Data

- Customer Key
- Σ Index
- Order Date
- Σ Order Line Item
- Order Number
- Σ Order Quantity
- Product Key
- Stock Date

Edit relationship

Select tables and columns that are related.

Sales Data

Order Date	Stock Date	Order Number	Product Key	Customer Key	Territory Key	Order Line Item	C
7/5/2020	6/3/2020	SO46718		360	12570	9	1
7/7/2020	4/22/2020	SO46736		360	12341	9	1
7/12/2020	5/5/2020	SO46776		360	12356	9	1

Calendar Lookup

Date	Day Name	Start of Week	Start of Month	Month Name	Start of Year	Year
1/1/2020	Wednesday	12/29/2019	1/1/2020	January	1/1/2020	2020
1/2/2020	Thursday	12/29/2019	1/1/2020	January	1/1/2020	2020
1/3/2020	Friday	12/29/2019	1/1/2020	January	1/1/2020	2020

Cardinality: Many to one (*:1)
Cross filter direction: Single
☒ Make this relationship active
☐ Assume referential integrity

OK Cancel

INACTIVE RELATIONSHIPS

The screenshot shows the Power BI interface with the 'Calendar ...' table selected. The 'Sales Data' table is also visible. A relationship is established between 'Order Date' in 'Sales Data' and 'Date' in 'Calendar ...'. The relationship is inactive, as indicated by the 'Make this relationship active' checkbox being unchecked in the 'Edit relationship' dialog.

Calendar ...

- Date
- Day Name
- Month Name
- Start of Month
- Start of Week
- Start of Year
- Σ Year

Sales Data

- Customer Key
- Σ Index
- Order Date
- Σ Order Line Item
- Order Number
- Σ Order Quantity
- Product Key
- Stock Date

Edit relationship

Select tables and columns that are related.

Sales Data

Order Date	Stock Date	Order Number	Product Key	Customer Key	Territory Key	Order Line Item	C
7/5/2020	6/3/2020	SO46718		360	12570	9	1
7/7/2020	4/22/2020	SO46736		360	12341	9	1
7/12/2020	5/5/2020	SO46776		360	12356	9	1

Calendar Lookup

Date	Day Name	Start of Week	Start of Month	Month Name	Start of Year	Year
1/1/2020	Wednesday	12/29/2019	1/1/2020	January	1/1/2020	2020
1/2/2020	Thursday	12/29/2019	1/1/2020	January	1/1/2020	2020
1/3/2020	Friday	12/29/2019	1/1/2020	January	1/1/2020	2020

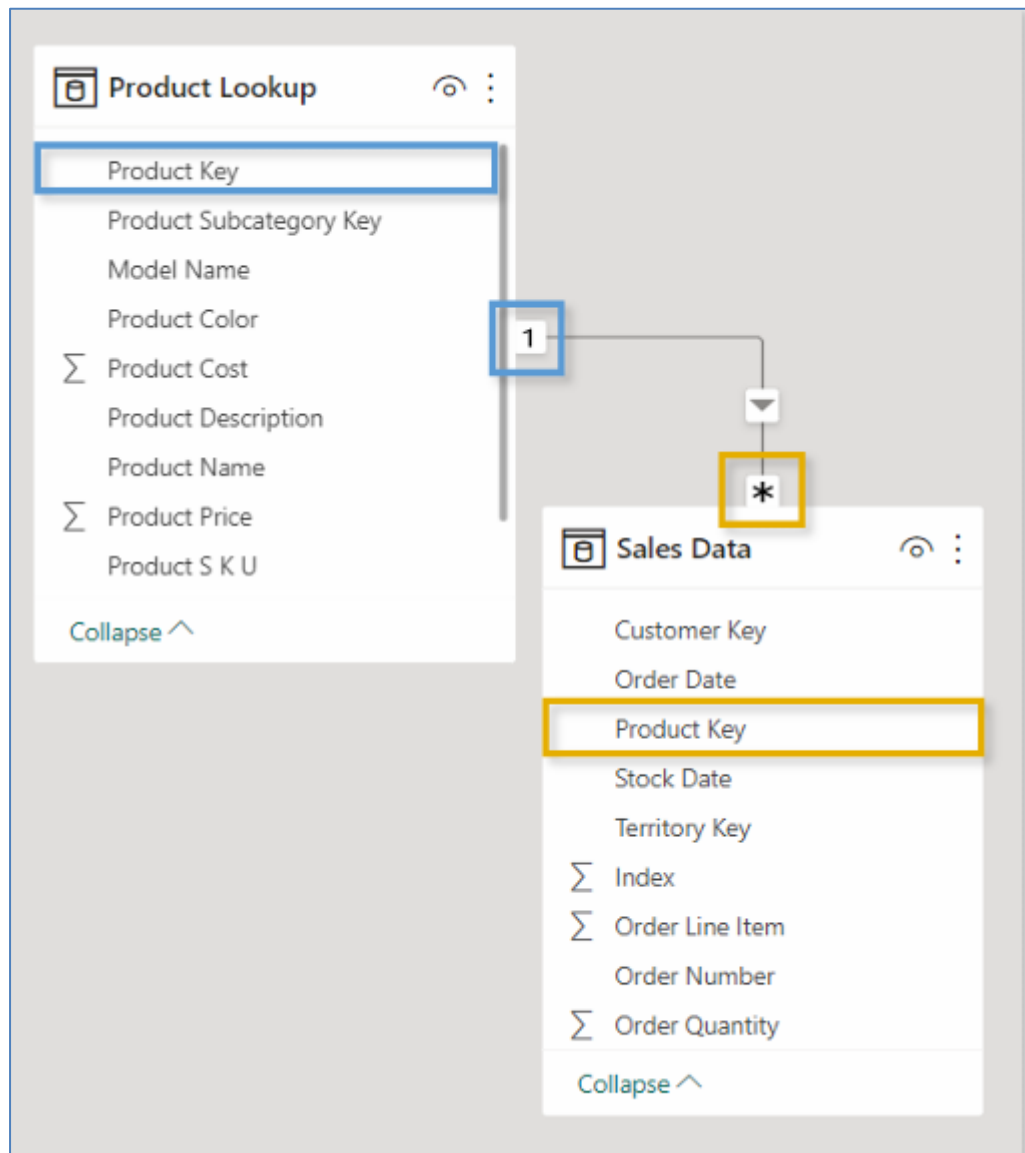
Cardinality: Many to one (*:1)
Cross filter direction: Single
☐ Make this relationship active
☐ Assume referential integrity

OK Cancel

RELATIONSHIP CARDINALITY

Cardinality refers to the uniqueness of values in a column

Ideally, all relationships in the data model should follow a **one-to-many** cardinality: **one** instance of each primary key, and **many** instances of each foreign key



EXAMPLE: ONE-TO-ONE CARDINALITY

product_id	product_name	product_sku
4	Washington Cream Soda	64412155747
5	Washington Diet Soda	85561191439
7	Washington Diet Cola	20191444754
8	Washington Orange Juice	89770532250

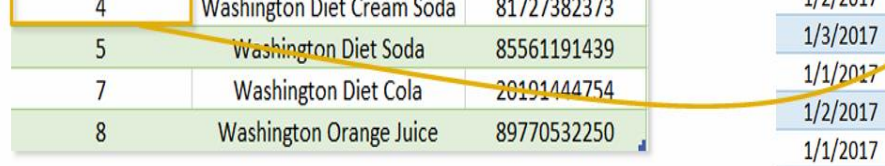
product_id	product_price
4	\$3.64
5	\$2.19
7	\$2.61
8	\$2.59



EXAMPLE: MANY-TO-MANY CARDINALITY

product_id	product_name	product_sku
4	Washington Cream Soda	64412155747
4	Washington Diet Cream Soda	81727382373
5	Washington Diet Soda	85561191439
7	Washington Diet Cola	20191444754
8	Washington Orange Juice	89770532250

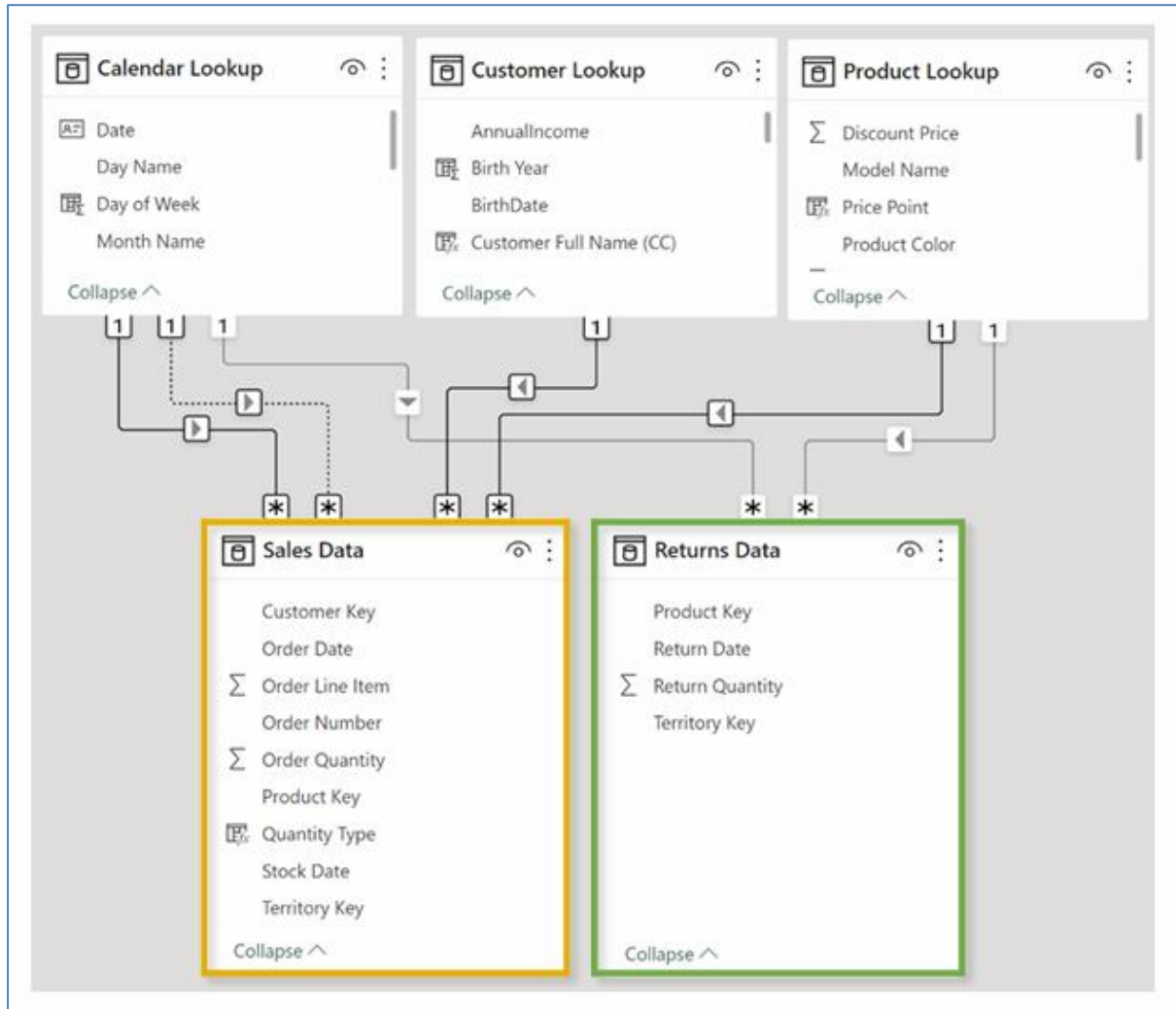
date	product_id	transactions
1/1/2017	4	12
1/2/2017	4	9
1/3/2017	4	11
1/1/2017	5	16
1/2/2017	5	19
1/1/2017	7	11



! This relationship has cardinality Many-Many. This should only be used if it is expected that neither column (product_id and product_id) contains unique values, and that the significantly different behavior of Many-many relationships is understood. [Learn more](#)

CONNECTING MULTIPLE FACT TABLES

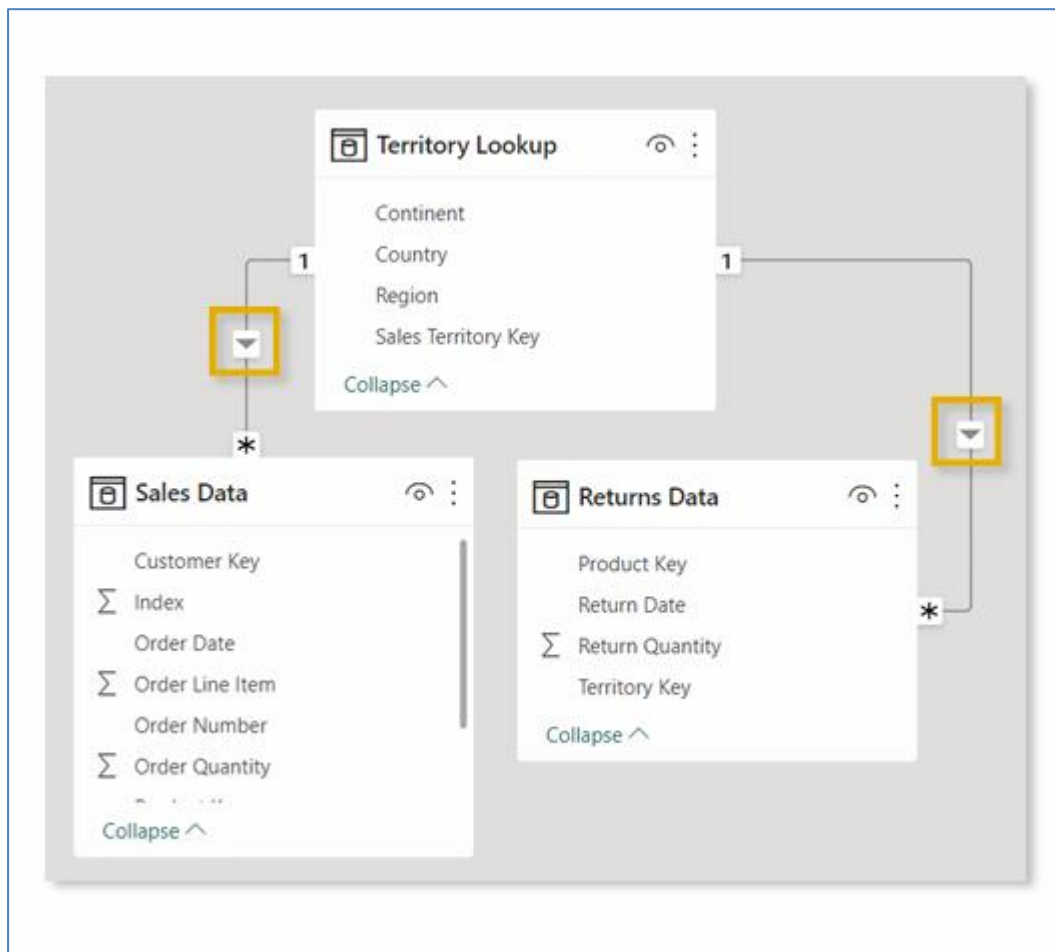
fact tables should connect through shared dimension tables, not directly to each other



FILTER CONTEXT & FLOW

The arrows show the filter direction, and point from the **one (1)** side of the relationship to the **many (*)** side

- When you filter a table, that filter context is passed to any related “**downstream**” tables, following the arrow’s direction
- Filter context CANNOT flow “**upstream**”



Arrange lookup tables above fact tables in your model as a visual reminder that filters always flow downstream.

FILTER FLOW:

TerritoryKey	OrderQuantity	ReturnQuantity
1	12,513	270
2	40	
3	30	
4	17,191	362
5	49	1
6	9,694	238
7	7,862	186
8	7,950	163
9	17,951	404
10	9,694	204
Total	84,174	1,828

Filtering by **Territory Lookup**[Territory Key]

TerritoryKey	OrderQuantity	ReturnQuantity
1	12,513	1,828
2	40	1,828
3	30	1,828
4	17,191	1,828
5	49	1,828
6	9,694	1,828
7	7,862	1,828
8	7,950	1,828
9	17,951	1,828
10	9,694	1,828
Total	84,174	1,828

Filtering by **Sales Data**[Territory Key]

TerritoryKey	OrderQuantity	ReturnQuantity
1	84,174	270
4	84,174	362
5	84,174	1
6	84,174	238
7	84,174	186
8	84,174	163
9	84,174	404
10	84,174	204
Total	84,174	1,828

Filtering by **Returns Data**[Territory Key]

BI-DIRECTIONAL FILTERS

Edit relationship

Select tables and columns that are related.

Sales Data

Order Date	Stock Date	Order Number	Product Key	Customer Key	Territory Key	Qty
Sunday, July 5, 2020	Wednesday, June 3, 2020	SO46718	360	12570	9	
Tuesday, July 7, 2020	Wednesday, April 22, 2020	SO46736	360	12341	9	
Sunday, July 12, 2020	Tuesday, May 5, 2020	SO46776	360	12356	9	

Territory Lookup

Region	Country	Continent	Sales Territory Key
Northwest	United States	North America	1
Northeast	United States	North America	2
Central	United States	North America	3

Cardinality: Many to one (*:1)

☒ Make this relationship active

☐ Assume referential integrity

☐ Apply security filter in both directions

Cross filter direction: **Both**

OK Cancel

The diagram illustrates the relationship between three tables: Sales Data, Returns Data, and Territory Lookup. Sales Data is connected to Territory Lookup (Sales Territory Key) with a cardinality of 1 to *. Returns Data is connected to Territory Lookup (Territory Key) with a cardinality of 1 to *. The cross-filter direction is set to 'Both', allowing filter context to flow in both directions between the Sales and Returns tables through the Territory Lookup table.

Properties

Relationship

Table: Sales Data Column: Territory Key

Cardinality: Many to one (*:1)

Table: Territory Lookup Column: Sales Territory Key

Make this relationship active: Yes

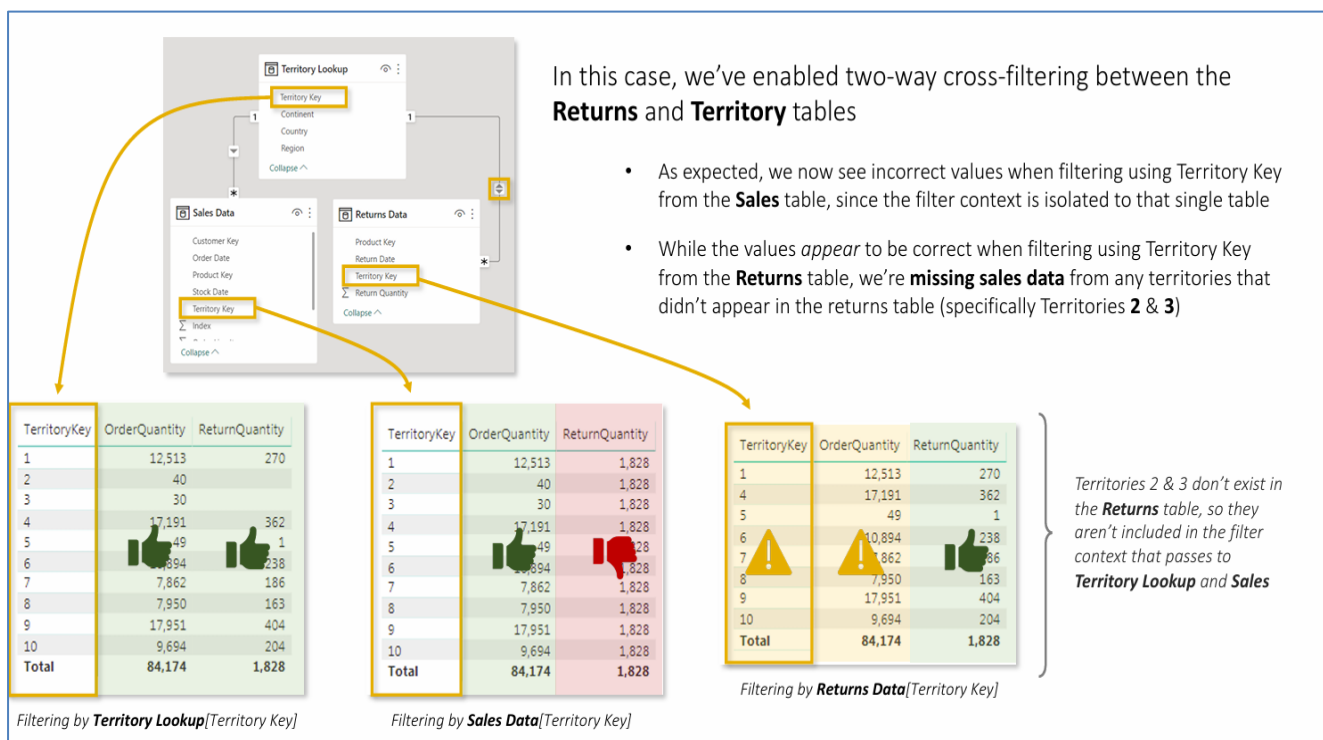
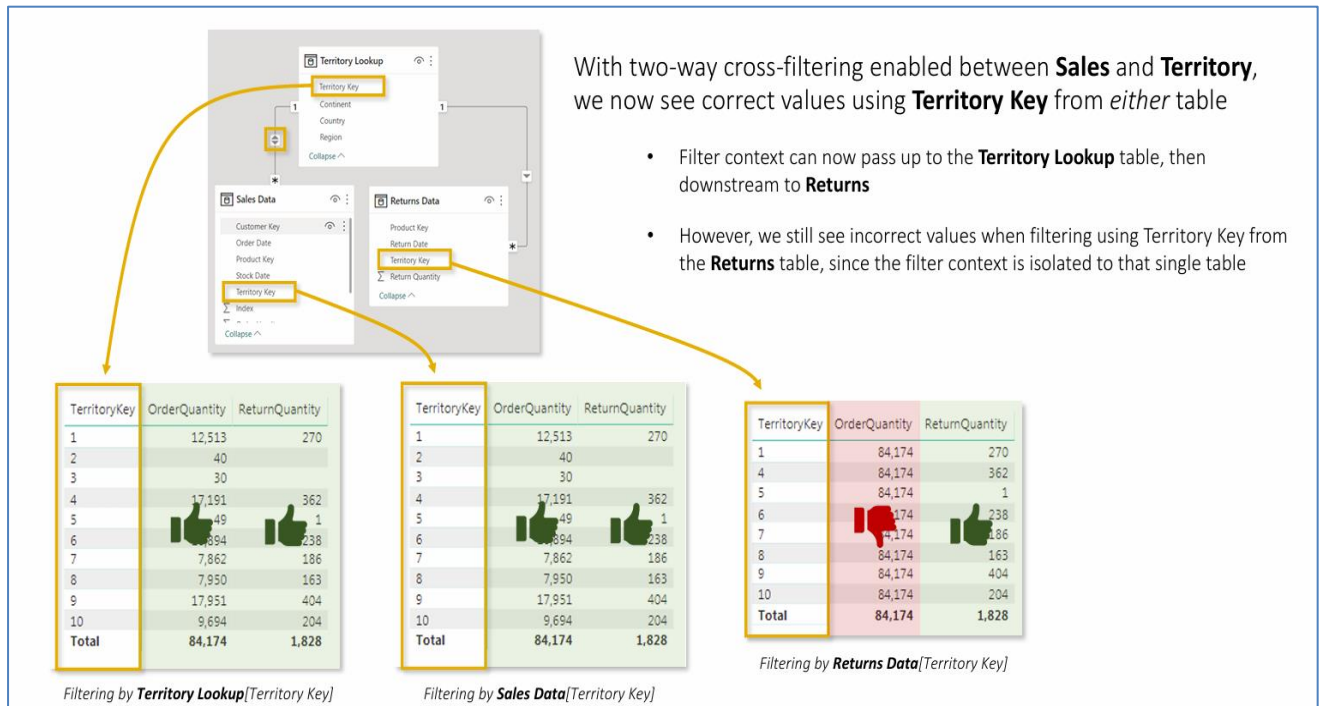
Cross filter direction: **Both**

Apply security filter in both directions: No

Updating the **cross-filter direction** from **Single** to **Both** allows filter context to flow in either direction

- In this example, filters applied to the **Sales** table can pass up to the **Territory Lookup** table, then down to **Returns**

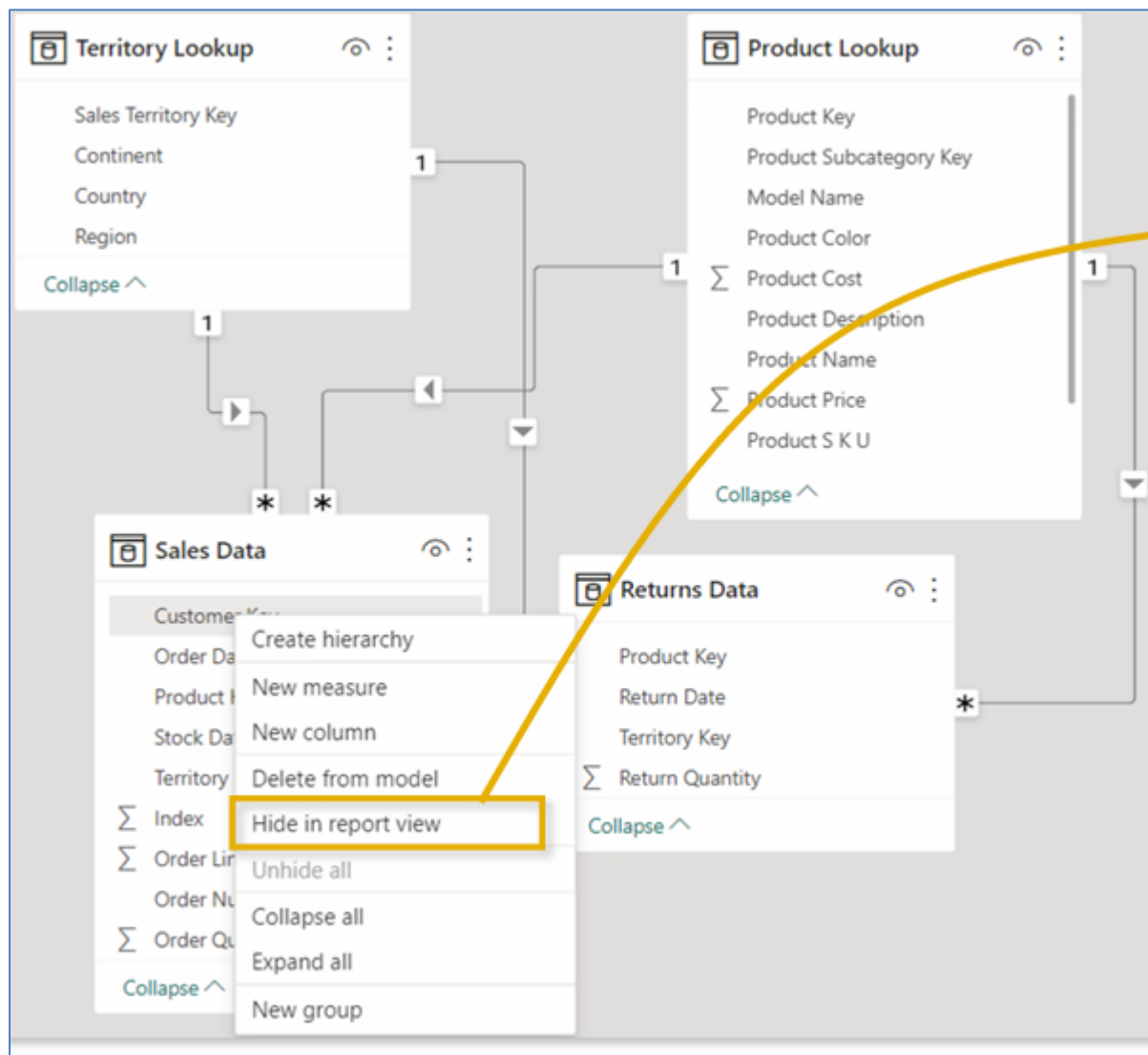
BI-DIRECTIONAL FILTER - Example



HIDING FIELDS

Hide in Report View makes fields inaccessible from the Report tab, but still available in Data and Model views

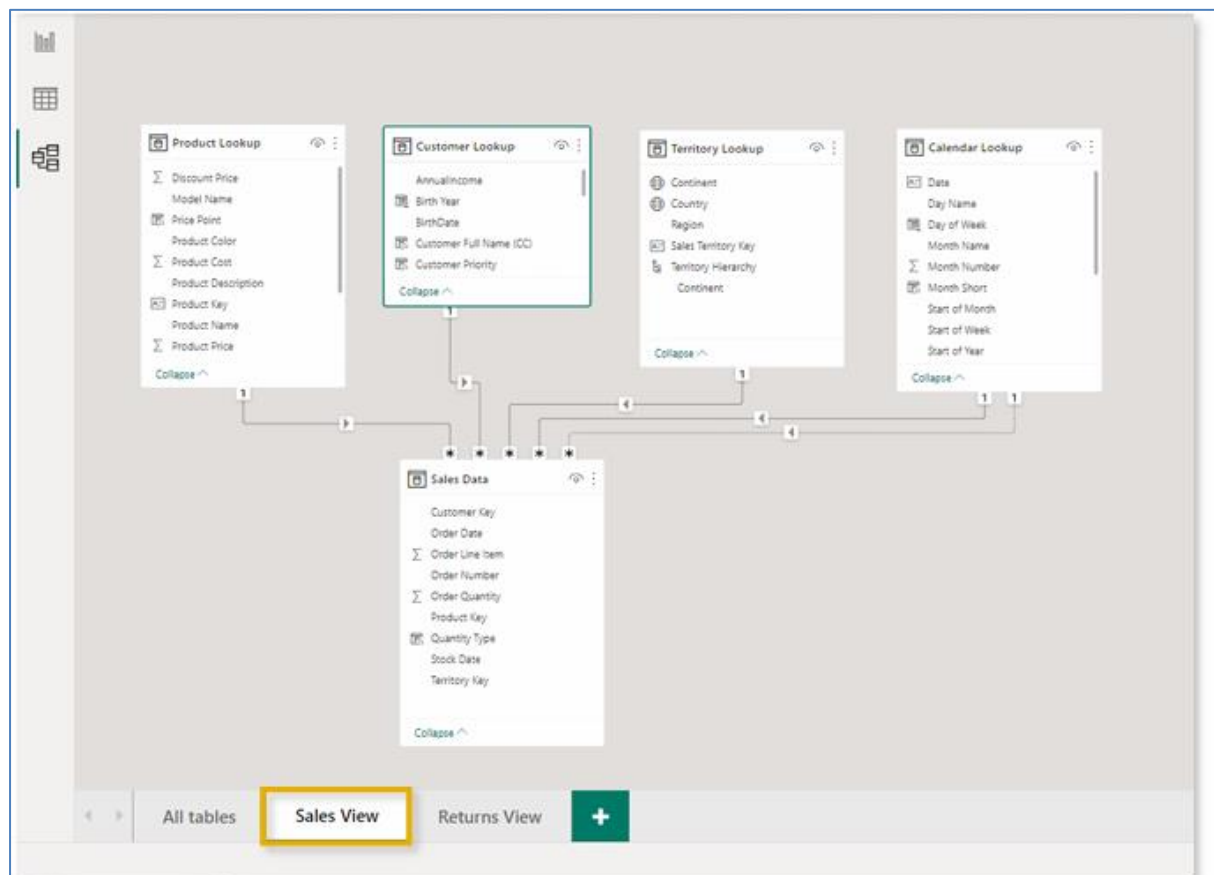
- This can be controlled by right-clicking a field in the Data or Model view, or by selecting “Is hidden” in the Properties pane
- This is commonly used to prevent users from filtering using invalid fields, reduce clutter, or to hide irrelevant metrics from view

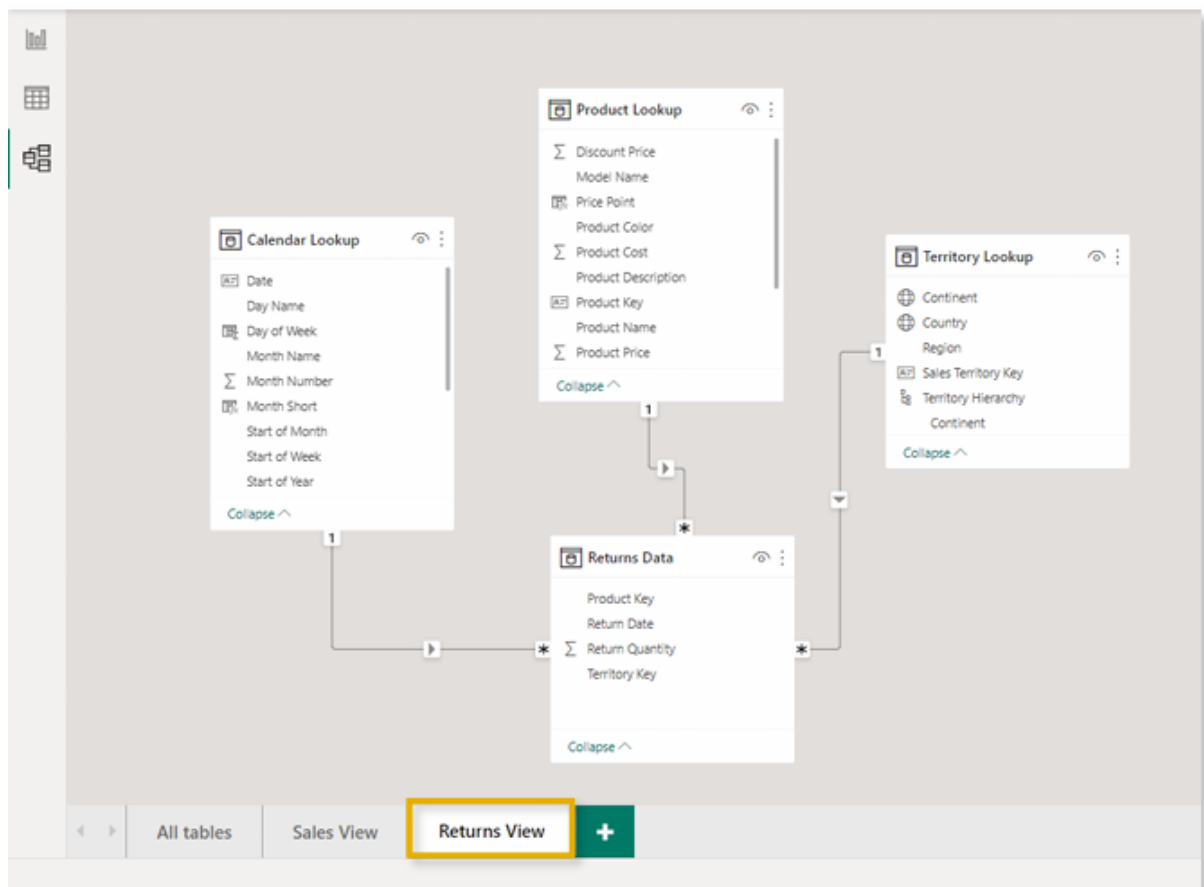


MODEL LAYOUTS

Model layouts allow you to create custom views to show specific portions of large, complex models

- Here we've created a Sales View displaying only tables related to sales, and a Returns View displaying only tables related to returns (Note: this doesn't actually create duplicate tables)





DATA FORMATS & CATEGORIES

File Home Help External Tools **Table tools** Column tools

Name Country

Data type Text

Format Text

\$ % % Auto

Summarization Don't summarize

Data category Country

Sort by column Sort

Structure

Formatting

Region	Country	Continent	Sales Territory Key
Northwest	United States	North America	1
Northeast	United States	North America	2
Central	United States	North America	3
Southwest	United States	North America	4
Southeast	United States	North America	5
Canada	Canada	North America	6
France	France	Europe	7
Germany	Germany	Europe	8
Australia	Australia	Pacific	9
United Kingdom	United Kingdom	Europe	10

Uncategorized

Address

Place

City

County

State or Province

Postal code

Country

Continent

Latitude

Longitude

Web URL

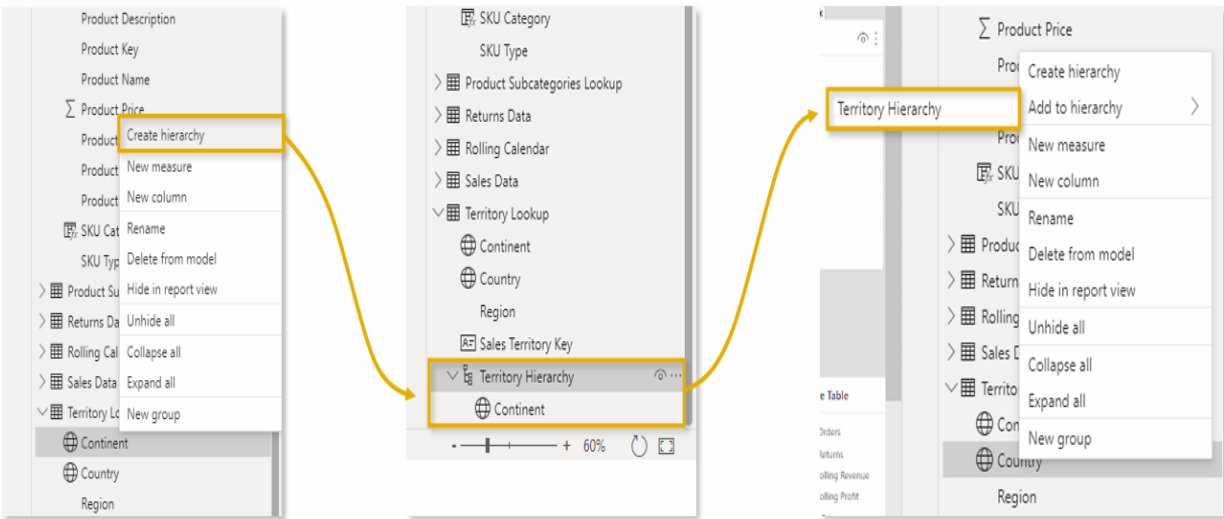
Image URL

Barcode

HIERARCHIES

Hierarchies are groups of columns that reflect multiple levels of granularity

- For example, a Geography hierarchy might include Country, State and City fields
- Hierarchies are treated as a single item in tables and reports, allowing users to “drill up” and “drill down” through each level



In the **Data** pane, right-click a field and select **Create hierarchy**

This hierarchy contains “Continent”, and is named “**Territory Hierarchy**”

Right-click another field (like “Country”) and select **Add to Hierarchy** (or drag it in!)